Stress Assignment without Morphology in European Portuguese: The Qualityless Vowel /0/ and the Metrically Marked Morpheme

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Abstract:

This paper offers an Optimality-Theoretic reinterpretation and synthesis of the author's earlier Japanese works — Makino (2010, 2012) — on Portuguese stress patterns, focusing exclusively on the theoretical essentials. We propose a **prosody-only account of European Portuguese word stress**. The system assumes a **qualityless vowel /0/ carrying one mora** in underlying representations, employed by specific morphemes when required. The default stress pattern arises from right-aligned, iterative binary trochaic footing, governed by the ranking *CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN; TROCHEE; ALIGN-R(Ft, PrWd); ALIGN-R(HdFt, PrWd). A small class of **metrically marked morphemes — MMMs** — triggers penultimate anchoring only when their own final vowel serves as the nucleus of the penultimate syllable of the prosodic word, formalised as ALIGN-R(Ft, σ_{final}_MMM) ≫ ALIGN-R(Ft, PrWd).

Metrification is computed strictly from phonological information — syllabification with Ç and syllable weight — together with a single piece of non-categorial lexical information, namely the MMM condition. No reference to syntactic categories or morphological bracketing is made at any point in the stress assignment stratum. This approach therefore removes morphology from stress computation proper, attributes apparent exceptions to /0/ and MMMs, and unifies items such as *contadora*, *contador*, *falávamos*, *falava*, *académico* and *após* under one minimal mechanism.

Abbreviations

PrWd = prosodic word; Ft = foot; HdFt = head foot; MMM = metrically marked morpheme; /0/ = qualityless vowel with one mora; C̩ = consonantal syllable nucleus that arises from the association of a consonant with the stray mora contributed by /0/; σ_{final}_MMM = the syllable whose nucleus is the final vowel of an MMM.

1. The qualityless vowel /0/

A **qualityless** vowel **/0/** that carries **one mora** in underlying representations is assumed — e.g. *contador* /kont-a-dor-**0**/ (cf. *contadora* /kont-a-dor-**a**/); *jacaré* /ʒakarɛ-**0**/; *trabalhar* /trabaʎ-a-r**0**/; *trabalharmos* /trabaʎ-a-r**0**-mos/; *após* /apɔs**0**/; *até* /atɛ**0**/.

2. Postulation of the MMM

It is also assumes that there exists a **classe of metrically marked morphemes (MMMs)** that are lexically specified to require **iterative binary footing from the penultimate syllable of the PrWd**, but **only when their own final vowel serves as the nucleus of that penultimate syllable** — e.g. *numer-* /numɛr-/ in *número* /**numɛr**-o/; *-ic-* /-ik-/ in *metafór**ico*** /mɛta-fɔr-**ik**-o/; *-vel* /-vel/ in *agradá**vel*** /a-grad-a-**vel**-0/; *-va-* /-va-/ in *trabalhá**va**mos* /trabaʎ-a-**va**-mos/; *-r-* /-r0-/ in *trabalha**r**mos* /trabaʎ-a-**r0**-mos/. Whether a morpheme belongs to the class of MMMs or not is independent of its categorical information.

3. Unmarked metrification

Regardless of syntactic category or morphological structure, **iterative formation of trochaic binary feet proceeds from the final syllable of the PrWd towards the word-initial edge**, with the **rightmost foot** serving as the head and bearing **primary stress**:

engorduramento /en-gord-ur-a-ment-o/ → en.gor.du.ra.men.to → (ˌen.gor)(ˌdu.ra)(ˈmen.to) → [ˌẽ.guɾˌðu.ɾɐˈmẽː.tu]
contador /kont-a-dor-0/ → kon.ta.do.r̩ → (ˌkon.ta)(ˈdo.r̩) → [ˌkõ.tɐˈðoːɾ]
contadora /kont-a-dor-a/ → kon.ta.do.ra → (ˌkon.ta)(ˈdo.ra) → [ˌkõ.tɐˈðoː.ɾɐ]
jacaré /ʒakarɛ-0/ → ʒa.ka.rɛ.0 → (ʒa.ka)(rɛ.0) → [ˌʒɐ.kɐˈɾɛː]
trabalhamos /trabaʎ-a-mos/ → tra.ba.ʎa.mos → (ˌtra.ba)(ˈʎa.mos) → [ˌtɾɐ.βɐˈʎɐː.muʃ]
trabalhar /trabaʎ-a-r0/ → tra.ba.ʎa.r̩ → (ˌtra.ba)(ˈʎa.r̩) → [ˌtɾɐ.βɐˈʎaːɾ]

This suggests the following constraints: **FTBIN; TROCHEE; ALIGN-R(Ft, PrWd); ALIGN-R(HdFt, PrWd)**.

4. Marked metrification

If the **final vowel of an MMM** serves as the **nucleus of the penultimate syllable of the PrWd**, **iterative formation of trochaic binary feet** proceeds **from that penultimate syllable** — rather than from the final syllable — with the **rightmost foot** serving as the head foot and bearing primary stress; the final syllable is excluded from metrification and remains unstressed:

***númer**o* /**numɛr**-o/ → **nu**.**mɛ**.**r**o → (ˈ**nu**.**mɛ**)**r**o → [ˈnuː.mɨ.ɾu]
euro /**euro**-0/ → **eu̯**.**ro**.0 → (ˈ**eu̯**.**ro**)0 → [ˈeu̯.ɾɔ]
*académ**ico*** /akadɛm-**ik**-o/ → a.ka.dɛ.m**i**.**k**o → (ˌa.ka)(ˈdɛ.m**i**)**k**o → [ˌɐ.kɐˈðɛː.mi.ku]
*agradá**vel*** /a-grad-a-**vel**-0/ → a.gra.da.**ve**.**l̩** → (ˌa.gra)(ˈda.**ve**)**l̩** → [ˌɐ.ɣɾɐˈðaː.vɛɫ]
*trabalhá**va**mos* /trabaʎ-a-**va**-mos/ → tra.ba.ʎa.**va**.mos → (ˌtra.ba)(ˈʎa.**va**)mos
→ [ˌtɾɐ.βɐˈʎaː.vɐ.muʃ]
*trabalha**r**mos* /trabaʎ-a-**r0**-mos/ → tra.ba.ʎa.**r̩**.mos → (ˌtra.ba)(ˈʎa.**r̩**)mos → [ˌtɾɐ.βɐˈʎaɾ.muʃ]

This suggests the ranking: **ALIGN-R(Ft, σ_{final}_MMM) ≫ ALIGN-R(Ft, PrWd)**. By contrast, if the final vowel of an MMM does not serve as the nucleus of the penultimate syllable of the PrWd, unmarked metrification in Section 3 takes place:

***numer**oso* /**numɛr**-oz-o/ → **nu.mɛ.r**o.zo → (ˌ**nu.mɛ**)(ˈ**r**o.zo) → [ˌnu.mɨˈɾoː.zu]
*atoni**ci**dade* /a-tɔn-**ik**-i-dad-e/ → a.tɔ.n**i.s**i.da.de → (ˌa.tɔ)(ˌ**ni.s**i)(ˈda.de) → [ˌɐ.tuˌni.siˈ ðaː.ðɨ]
*trabalha**va*** /trabaʎ-a-**va/** → tra.ba.ʎa.**va** → (ˌtra.ba)(ˈʎa.**va**) → [ˌtɾɐ.βɐˈʎaː.vɐ]
*trabalha**r*** /trabaʎ-a-**r0/** → tra.ba.ʎa.**r̩** → (ˌtra.ba)(ˈʎa.**r̩**) → [ˌtɾɐ.βɐˈʎaːɾ]

5. Stress clash avoidance (provisional)

When the metrification domain contains an **odd** number of syllables, to avoid a stress clash between a potential **word-initial unary foot** and the following **binary foot**, metrification proceeds as follows. These generalisations are **provisional** and call for further investigation.

5.1 Five or more syllables: only the leftmost foot is ternary

Only the **leftmost** foot is **ternary**; **subsequent** feet remain **binary** trochees:

papelaria /papel-ari-a/ → pa.pe.la.ri.a → (ˌ**pa.pe.la**)(ˈri.a) → [ˌpɐ.pɨ.lɐˈðaː.ðɨ]

computado /konput-a-dor-0/ → kon.pu.ta.do.ɾ̥ → (ˌ**kon.pu.ta**)(ˈdo.ɾ̥) → [ˌkõ.pu.tɐˈðoːɾ]
entendimento /entend-i-ment-o/ → en.ten.di.men.to → (ˌ**en.ten.di**)(ˈmẽ.to) → [ˌẽ.tẽ.diˈẽː.tu]

Winner #(ˌ*)...#:** NO *CRASH violation + **NO PARSE(σ) violation** + **ONE FTBIN violation**
Loser # * (ˌ)...#**: NO *CRASH violation + **ONE PARSE(σ) violation** + **NO FTBIN violation**
∴ **PARSE(σ) ≫ FTBIN**

Winner #(ˌ*)...#**: **NO *CRASH violation** + NO PARSE(σ) violation + ONE FTBIN violation
Loser #(ˌ*)(ˌ)...#**: **ONE *CRASH violation** + NO PARSE(σ) violation + ONE FTBIN violation
∴ ***CRASH**: tie-breaking

This sugges the following constraints: **PARSE(σ) ≫ FTBIN**; ***CRASH**

5.2 Initial L-H exception: initial light syllable unfooted

If a **light initial syllable is immediately followed by a heavy syllable**, the initial light syllable remains **unfooted/extrametrical**:

levantamento /levant-a-ment-o/ → **le**.van.ta.men.to → **le**(ˌvan.ta)(ˈmen.to) → [lɨˌvɐ̃.taˈmẽː.tu]

Note that optional unary footing of the initial light syllable is not excluded, as in *levantamento* /levant-a-ment-o/ → **le**.van.ta.men.to → (ˌ**le**)(ˌvan.ta)(ˈmen.to) → [ˌlɨˌvɐ̃.taˈmẽː.tu].

Winner # L (ˌH.σ)...#:
NO *CRASH violation + **ONE PARSE(σ) violation** + NO FTBIN violation + **NO #WSP violation**
Loser # (ˌL.H.σ)...#:
NO *CRASH violation + **NO PARSE(σ) violation** + ONE FTBIN violation + **ONE #WSP violation**
⇒ **#WSP ≫ PARSE(σ)**: since PARSE(σ) ≫ FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

Winner # L (ˌH.σ)...#:
NO *CRASH violation + **ONE** PARSE(σ) violation + **NO** FTBIN violation + **NO** #WSP violation
Loser # (ˌL)(ˌH.σ)...#:
ONE *CRASH violation + **NO** PARSE(σ) violation + **ONE** FTBIN violation + **NO** #WSP violation
∴ ***CRASH** decides: tie-breaking
This sugges the following constraints: **WSP ≫ PARSE(σ) ≫ FTBIN**; ***CRASH**

5.3 Three syllables: initial syllable unfooted

The **initial syllable is not footed**:

pessoa /peso-a/ → **pe**.so.a → **pe**(ˈso.a) → [pɨˈsoː.ɐ]
rapaz /rapas-0-s/ → **ra**.pa.ș → **ra**(ˈpa.ș) → [ʀɐˈpaːʃ]
pequeno /peken-o/ → **pe**.ke.no → **pe**(ˈke.no) → [pɨˈkeː.nu]
levar /lɛv-a-r0/ → **lɛ**.va.ṛ → **lɛ**(ˈva.ṛ) → [lɨˈvaːɾ]
até /atɛ0/ → **a**.tɛ.0 → **a**(ˈtɛ.0) → [ɐˈtɛː]
após /apɔs0/ → **a**.pɔ.ș → **a**(ˈpɔ.ș) → [ɐˈpɔːʃ]

Note that optional unary footing of the initial syllable, as in *levar* /lɛv-a-r0/ → lɛ.va.ṛ → (lɛ)(ˈva.ṛ) → [ˌlɨˈvaːɾ], is not excluded.

Winner # * (ˈ)#: NO *CRASH violation + ONE PARSE(σ) violation + NO** FTBIN violation
Loser #(ˌ*)(ˈ)#: ONE *CRASH violation + NO PARSE(σ) violation + ONE** FTBIN violation
∴ ***CRASH ≫ PARSE(σ)**:since PARSE(σ) ≫ FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This sugges the following constraints: ***CRASH, WSP ≫ PARSE(σ) ≫ FTBIN**

6. The constraint set (provisional)

From the discussion in Sections 3, 4 and 5, the following set of constraints is inferred:

***CRASH, WSP ≫ PARSE(σ) ≫ FTBIN;**
TROCHEE;
ALIGN-R(Ft, σ_{final}_MMM) ≫ ALIGN-R(Ft, PrWd);
ALIGN-R(HdFt, PrWd)

7. Conclusion

Thus, Portuguese stress can be entirely reduced to **qualityless vowel /0/ with one mora + metrically marked morphemes + general stress theory**, eliminating the need for morphologically-driven stress assignment.

Reference

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